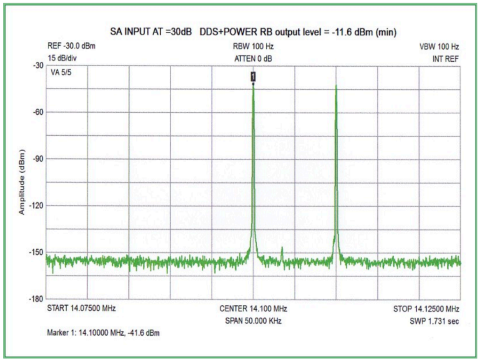
1)



a) a= N= -155 dBm

Tcondizione nel tempo dispositivi

$N = -174\ dBm/Hz + 10\log(RBW)$

$= -154\ dBm$

b) S_1 ex $SNR \approx max - min = -45 + 155 = 110$

S_2 ex $SNR \approx max - min = -45 + 155 = 110$

c) dispositivo i 1 Hz bandwidth

RBW = 10 Hz

$$N_0 = N - 10\log\left(\frac{RBW}{1\ Hz}\right)$$

$= -175\ dBm/Hz$

d) $\Delta N = 10\log\frac{105}{12} = 30dB$

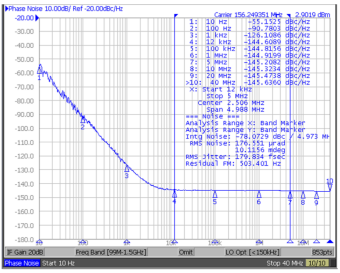
2) f = 156.25 MHz

$f_{phase\ detector} = 5\ GHz$

Hanno mano ganga generata dous

$N = \frac{5\ GHz}{156.25\ MHz} = 32$

Altri multipliers PLLs generano nel 32



De blir forsterkningen økt med

$20 \log(32) = 30.1$ dB

Forsterkeren har høypasfilteren

$\angle(100\ Hz) = -90.78$ dB c/Hz

$\angle(1\ kHz) = -126.11$ dBc/Hz

$\angle_{ut}(100) = -90.78 + 30.1 = -60.7$

$\angle_{ut}(1k) = -126.11 + 30.1 = -96$

3) Forbedring: spektralanalysen

- Bruke bandpass LC eller RC
- overblingskondensator
- sikring for å unngå inn utstråling selv om komponenten
- lineær-regulator